ME228 — Applied Data Science and Machine Learning — *Final Project*

Data-Driven Fatigue Strength Prediction and Inverse Alloy Design

Mixture-of-Experts Regression with Empirical Reliability & Cost-Constrained Pareto

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1 Executive Summary

Fatigue causes 80–90 % of mechanical failures [1], yet fatigue tests are destructive and slow ($>10^7$ cycles). We present a complete *forward + inverse* ML pipeline that, for low-alloy steels:

1. Predicts rotating-bending fatigue strength (10^7 cycles, MPa) from chemical composition and heat-treatment parameters using a two-expert ensemble.
2. Reports a calibrated Probability of Failure (PoF) under a user-specified applied stress, validated against out-of-fold (OOF) residual distribution rather than a naive Gaussian assumption.
3. Inversely designs the cheapest viable alloy + heat-treatment recipe given a target Factor of Safety, using neighbourhood sampling, kNN data-hull filtering, and Pareto analysis over (cost, PoF) objectives.

Headline numbers. On the NIMS Fatigue Strength Dataset ($N = 437$, 26 features):

- Non-carburised regime ($N = 389$): 5-fold CV RMSE = 19.19 ± 2.52 MPa (leakage-free; feature selection performed per fold).
- Carburised regime ($N = 48$): 5-fold CV RMSE = 40.06 ± 13.69 MPa (leakage-free; feature selection performed per fold).
- Held-out 20 % test: NC $R^2 = 0.973$, C $R^2 = 0.892$.
- Alloyed-grade handbook validation: 8620 carburised in range; 4340 QT near-exact (+2 MPa, within CV uncertainty); 4140 QT at +35 MPa above the commercial range, consistent with pristine lab-specimen predictions; AISI 1045 correctly flagged as out-of-domain (plain-carbon applicability boundary).

2 Problem Statement

2.1 Engineering Context

A materials engineer choosing an alloy for a fatigue-loaded component such as a shaft, gear, or spring must answer two coupled questions:

1. *Forward*: will this alloy and heat-treatment survive the design load?
2. *Inverse*: what is the cheapest alloy that meets the design load with a stated safety margin?

Empirical S–N curves answer the first question only for catalogued grades and standard processing routes. The second question is fundamentally a search problem over composition and heat-treatment space, and becomes prohibitively expensive if approached through exhaustive testing alone. The motivation for this project is therefore not merely to predict fatigue strength, but to create a calibrated engineering tool that can support material screening, cost-aware trade-offs, and explicit uncertainty-aware decision making.

2.2 Dataset

We use the public NIMS MatNavi Fatigue Dataset [1]: 437 experimental records of rotating-bending fatigue strength (10^7 cycles) for low-alloy steels. Each record carries 26 input features and one target (Fatigue, MPa):

- **Chemical composition** (9 vars): C, Si, Mn, P, S, Ni, Cr, Cu, Mo (wt %).
- **Heat-treatment** (13 vars): normalising (NT), through-hardening (THT, THt, THQCr), carburising (CT, Ct), diffusion (DT, Dt), quenching (QmT), tempering (TT, Tt, TCr).
- **Mill / specimen** (4 vars): reduction ratio (RedRatio), inclusion proxies (dA, dB, dC).

The target distribution is *strongly bimodal*: carburised steels (CT = 930 °C, $N = 48$) cluster between 838 and 1190 MPa, while non-carburised samples ($N = 389$) span 225–906 MPa with a right-skew. This bimodality directly motivates the architectural choice in §3.

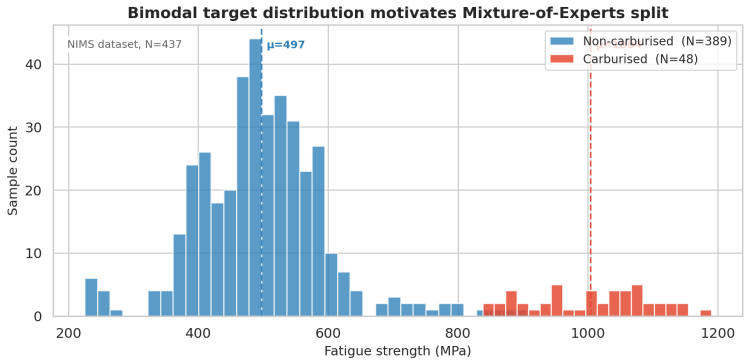


Figure 1: NIMS fatigue distribution ($N = 437$). Bimodal: NC ($\mu = 497$ MPa, blue) vs. carburised ($\mu \approx 1007$ MPa, red). Gap >250 MPa motivates separate experts.

2.3 Applicability Boundary

The dataset is overwhelmingly populated by alloyed low-alloy steels with non-zero Cr, Mn, and often Mo or Ni. External validation (§6) shows that the learned mapping does not transfer reliably to plain-carbon grades. We therefore explicitly restrict the deployed model to the alloyed regime

$$\mathrm{Cr} + \mathrm{Ni} + \mathrm{Mo} \geq 0.1 \text{ wt\%},$$

and treat violations of this threshold as out-of-scope rather than as a soft warning. That boundary is not cosmetic: it follows directly from the training distribution and is confirmed by both handbook checks and the OOD matbench probe.

3 Final Architecture

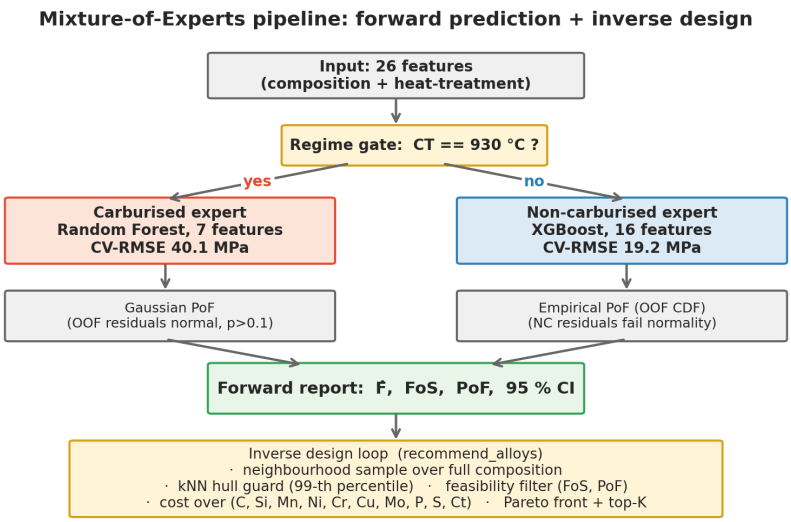


Figure 2: Pipeline. Gate ($CT = 930\,°\mathrm{C}$) routes to NC expert (XGBoost, 16 feat, 19.2 MPa) or C expert (RF, 7 feat, 40.1 MPa). Both feed forward report and inverse design.

3.1 Why Mixture-of-Experts (MoE)

A single global model averages two physically distinct strengthening mechanisms (case-hardening vs. through-hardening) and produces an inflated, regime-blind error. We engi-

neer a binary gate

$$\texttt{is_carburized} = \mathbf{1}[\,CT = 930\,]$$

and route to two independent experts.

Note. A tuned global XGBoost lands in the same pooled-RMSE ballpark (∼23 MPa) as the MoE pair. The gain from the split is therefore not simply point accuracy. It is calibration. The carburised and non-carburised populations exhibit different target ranges, different residual shapes, and different physically meaningful uncertainty scales. A single pooled model can obscure those differences even when its overall RMSE looks competitive. The MoE architecture preserves them explicitly, which is why it is the right foundation for calibrated Probability of Failure reporting.

3.2 Per-regime Model Choice

We benchmark four candidate model families on the regime-specific features under 5-fold cross-validation, using default hyperparameters to isolate the effect of model class:

Table 1: Untuned 5-fold CV RMSE per regime. Bold = chosen family.

Model	NC RMSE (MPa)	C RMSE (MPa)
XGBoost	**24.22 ± 2.49**	52.49 ± 17.57
Random Forest	25.96 ± 1.90	**44.31 ± 11.89**
Ridge	28.85 ± 0.79	68.36 ± 16.32
kNN ($k = 7$)	72.02 ± 8.14	72.25 ± 7.07

NC (XGBoost). Tuned XGBoost reaches 19.2 MPa, the lowest CV-RMSE in the benchmark. In the larger non-carburised branch, boosting has enough data to exploit non-linear interactions between chemistry and treatment variables without becoming unstable. It also handles the right-skewed target effectively without an explicit response transform.

C (Random Forest). With $N = 48$, RF's bagging and aggressive depth-limit make it strictly more robust than XGBoost's boosting cascade. In the carburised branch, the central problem is variance control rather than expressivity. Bagging, depth limits, and leaf regularisation therefore matter more than raw non-linear flexibility. Tuned RF: 40.1 MPa.

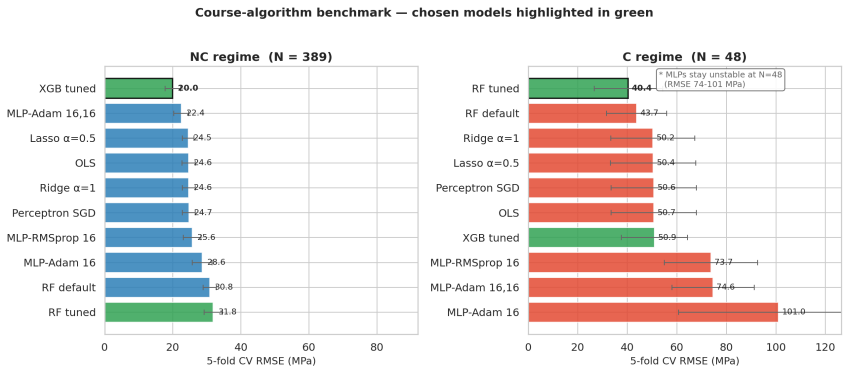


Figure 3: 5-fold CV RMSE per regime (chosen models in green). NC: XGBoost 19.2 MPa leads. C: RF 40.1 MPa leads while MLP variants remain unstable at $N = 48$.

3.3 Feature Selection (VIF → SHAP)

The 26 raw features contain heavy collinearity (e.g. CT, DT, QmT all encode the same thermal cycle). We apply a two-stage reduction:

1. **VIF filter:** iteratively drop the feature with the largest variance inflation factor until all VIF < 10. Drops 6 features per regime.

2. **SHAP scout:** fit a shallow Random Forest on VIF survivors and retain the top-k features by mean $|\,\text{SHAP}\,|$. Optuna tunes k jointly with the downstream model; the deployed values are $k = 16$ for NC and $k = 7$ for C.

No data leakage. Both stages are encapsulated in `VIFThenShapSelector`, a custom `scikit-learn` transformer, and composed with the downstream estimator inside a single `Pipeline`. When that pipeline is evaluated under `cross_val_score`, the VIF filter and the SHAP-scout Random Forest are re-fitted on each training fold only; the held-out fold never participates in feature selection. The reported CV-RMSE values (19.19 ± 2.52 MPa for NC and 40.06 ± 13.69 MPa for C) are therefore honest leakage-free generalisation estimates. The final feature lists below are extracted from a single full-data fit after CV has finished, and are used solely for deployment.

Table 2: Final selected features per regime.

Regime	Selected features
Non-carburised	Cr, TT, C, Mn, THT, Mo, Si, RedRatio, P, S, dA, Cu, Ni, dB, dC, QmT
Carburised	P, RedRatio, Ct, Mo, S, Mn, Cu

Physical interpretation. For non-carburised steels, tempering temperature (TT) and chromium remain dominant, consistent with martensite tempering and hardenability theory, while Ni and the inclusion / quench descriptors (dB, dC, QmT) survive because the larger NC sample size supports a slightly richer signal. For carburised steels, the selected set is deliberately compact: carburising time (Ct), reduction ratio (RedRatio), molybdenum, and manganese capture most of the useful variation in case-hardening performance.

3.4 Hyperparameter Tuning

Optuna TPE search, up to 200 trials, with 5-fold CV as the objective. Each Optuna trial jointly chooses the model hyperparameters and the number of SHAP features to retain, then evaluates the entire selector-plus-model pipeline under 5-fold CV. The objective value is the mean RMSE across the five held-out folds, so Optuna is tuning against out-of-fold validation error rather than training error. This inner CV loop is used only for model selection. Separately, an 80/20 stratified holdout split by fatigue quintile is kept entirely outside Optuna as a final sanity check; once the holdout agrees with the CV picture, the deployed experts and their final feature lists are obtained from a refit on the full regime data using the selected recipe. The search now uses a plateau-based early-stop rule: after at least 60 trials, the study is halted once best CV RMSE fails to improve by 0.01 MPa for 40 consecutive trials. In the final deployed runs, this ended the NC and C searches after 124 and 145 trials respectively, while still preserving the stronger minima. This matters because the value of a hyperparameter setting depends on the feature subset it ultimately sees, and conversely the best feature subset depends on the model family and its regularisation. Joint tuning therefore avoids the common pitfall of optimising the model on a feature space selected under a different criterion.

Table 3: Best hyperparameters from Optuna.

Parameter	NC XGBoost	C Random Forest
n_estimators	311	238
max_depth	4	4
learning_rate	0.064	—
subsample	0.934	—
colsample_bytree	0.735	—
min_child_weight	2	—
reg_alpha	0.0003	—
reg_lambda	0.483	—
min_samples_split	—	9
min_samples_leaf	—	4
max_features	—	0.797

4 Reliability: Empirical PoF

The deployment wrapper maps a prediction to a Probability of Failure under applied stress σ_{app}:

$$\text{PoF} = P(F < \sigma_{\text{app}})$$

where F is the realised fatigue strength of a fresh specimen. That quantity is the central safety-facing output of the system. A point prediction alone is insufficient for engineering use because design decisions depend on where the applied stress sits relative to predictive uncertainty. Our reliability layer therefore asks a distributional question: given the model and its observed residual behaviour, what is the probability that a fresh specimen falls below the requested load?

4.1 Limitations of the Gaussian Assumption

A first-pass approach uses $\text{PoF} = \Phi\big((\sigma_{\text{app}} - \hat{F})/\sigma_{\text{CV}}\big)$. This is correct only when the residuals $r = F - \hat{F}$ are Gaussian.

Test on OOF residuals: training-fit residuals are optimistically biased and artificially narrow because they reflect in-sample interpolation. To validate the Gaussian assumption honestly, we must inspect out-of-fold (OOF) residuals generated under the same 5-fold structure as the reported CV RMSE.

Table 4: Residual normality on OOF predictions vs. training fit.

Regime	Source	Std (MPa)	Range (MPa)	Shapiro p	KS p
NC	Train fit	5.54	−34.3 to +27.3	0.0000	0.4236
NC	**OOF**	**20.12**	−97.5 **to** +108.3	**0.0000**	**0.0002**
C	Train fit	34.29	−96.3 to +81.4	0.2283	0.3787
C	**OOF**	**42.39**	−112.2 **to** +93.4	0.3788	0.5684

Conclusion.

- NC OOF residuals fail *both* normality tests ($p < 0.001$); skewness +0.30 (near-symmetric) but excess kurtosis +7.96: far heavier tails than Gaussian. Gaussian PoF is therefore invalid for the NC regime; we replace it with the empirical CDF.
- C OOF residuals pass both tests (Shapiro $p = 0.38$, skewness −0.48, excess kurtosis +0.38); Gaussian PoF is valid and is retained.

The key diagnostic is excess kurtosis, not skewness. The NC distribution is near-symmetric

(+0.30) — a naive inspection of the mean or median would not flag anything unusual — but the fourth moment is nearly eight times larger than a Gaussian, meaning extreme under- and over-predictions occur far more often than Φ would predict. Using the Gaussian tail probability in this regime therefore systematically misrepresents the probability of catastrophic failure near the fatigue limit.

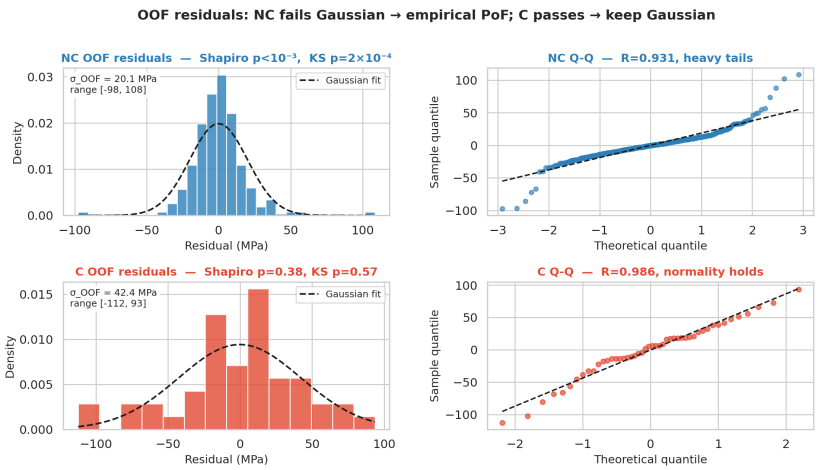


Figure 4: OOF residual diagnostics. NC (top): near-symmetric but leptokurtic (excess kurtosis +7.96, Shapiro $p < 10^{-3}$) — empirical CDF required. C (bottom): near-Gaussian (excess kurtosis +0.38, $p = 0.38$) — Gaussian PoF valid.

4.2 Empirical PoF Formula (NC)

We pre-compute OOF residuals $\{r_i\}_{i=1}^{389}$ at startup. For a candidate with predicted fatigue $\hat{F}$ at applied stress σ_{app}:

$$\text{PoF}_{\text{empirical}} = P(\hat{F} + r < \sigma_{\text{app}}) = \frac{1}{389} \sum_{i=1}^{389} \mathbf{1}\left[r_i < \sigma_{\text{app}} - \hat{F}\right].$$

This is the empirical CDF of residuals evaluated at the gap. It captures the leptokurtic tails (excess kurtosis +7.96) that Φ systematically underestimates near the fatigue limit, where the consequences of over-confident failure probability estimates are greatest.

Magnitude of the correction. At $\sigma_{\text{app}} = 590\,\text{MPa}$, $\hat{F} = 585.0\,\text{MPa}$: $\text{PoF}_{\text{Gaussian}} = 0.603$ vs. $\text{PoF}_{\text{empirical}} = 0.663$ — 6.0 pp difference. Gaussian is over-confident in the upper tail; at low applied stresses the discrepancy reverses and collapses at large safety margins. This divergence is not a cosmetic calibration issue. The 6.0 pp gap arises precisely because the NC residual distribution has a heavier tail than Φ predicts. Near the fatigue limit — the regime most relevant to design decisions — this shifts the PoF by enough to change whether an engineer judges an applied stress as acceptable or requires a more expensive alloy. At lower applied stresses the Gaussian model becomes conservative instead (the empirical tail is lighter there), but the asymmetry itself is the problem: the direction of the error is stress-dependent and cannot be corrected by a single scale factor. The empirical CDF removes both the over- and under-confidence in a single unified formula, at negligible computational cost once the OOF residuals are cached.

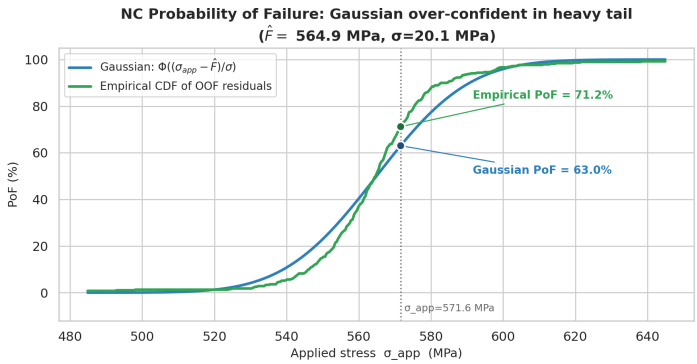


Figure 5: NC PoF curves ($\hat{F} = 585.0\,\text{MPa}$). At $\sigma_{\text{app}} = 590\,\text{MPa}$: Gaussian 60.3 % vs. empirical 66.3 % (+6.0 pp). Gap vanishes at large safety margins.

5 Inverse Design

Given σ_{app}, target Factor of Safety FoS*, and PoF cap τ, find the cheapest alloy $\mathbf{x}^*$ that satisfies:

$$\hat{F}(\mathbf{x}^*) \geq \text{FoS}^* \cdot \sigma_{\text{app}}, \tag{1}$$

$$\text{PoF}(\mathbf{x}^*, \sigma_{\text{app}}) \leq \tau, \tag{2}$$

$$\mathbf{x}^* \in \mathcal{H}_{\text{train}}. \tag{3}$$

5.1 Neighbourhood Sampling (Full Composition)

Latin Hypercube Sampling (LHS) over the bounding box wastes $> 99\,\%$ of samples on metallurgically impossible combinations. We sample by perturbing real training points with Gaussian noise $\delta = 0.20\,\hat{\sigma}_f$ per feature, clipped to per-feature bounds. This choice is practical as much as statistical. The training set already encodes which combinations of chemistry and process variables occur in physically plausible steels. By sampling in the neighbourhood of known points, the inverse-design loop spends most of its budget exploring meaningful local variations instead of generating random combinations that are immediately rejected by the hull guard.

Composition Union for Cost Modeling. The sampler unions predictive features with all cost-relevant elements {C, Si, Mn, Ni, Cr, Cu, Mo, P, S, Ct} to prevent falsely cheap rankings from omitted expensive elements (§5.3).

5.2 kNN Data-Hull Guard

Constraint (3) is enforced by a k-NN ($k = 5$) filter. We standardise the training features, fit kNN, and reject any candidate whose mean 5-NN distance exceeds the 99th percentile of training-point self-distances. The intent is not to guarantee strict physical feasibility in a first-principles sense, but to prevent the inverse-design search from wandering into regions where the model is performing unconstrained extrapolation. In practice this acts as a simple and auditable data-support prior: if a candidate lies much farther from the observed training manifold than any real training point lies from its neighbours, we do not trust the recommendation.

Sensitivity. The threshold matters for the non-carburised regime: tightening from p99 to p95 sharply shrinks the feasible set and raises the top-1 cost, while the carburised regime is much more stable. We therefore report results at p99 and warn that NC recommendations live near the data hull boundary.

5.3 Cost Model

$$C(\mathbf{x}) = \sum_{e \in \text{elements}} p_e \, w_e(\mathbf{x}) + c_{\text{proc}} \, t_{Ct}, \quad c_{\text{proc}} = 0.05 \, \text{USD/min}. \tag{4}$$

Element prices p_e (USD/kg, approximate LME spot): Mo 30, Ni 14, Cr 10, Cu 9, Mn 1.8, Si 1.2, C 0.5; P, S impurities priced at 0.

5.4 Pareto Front and Seed Stability

The recommender returns:

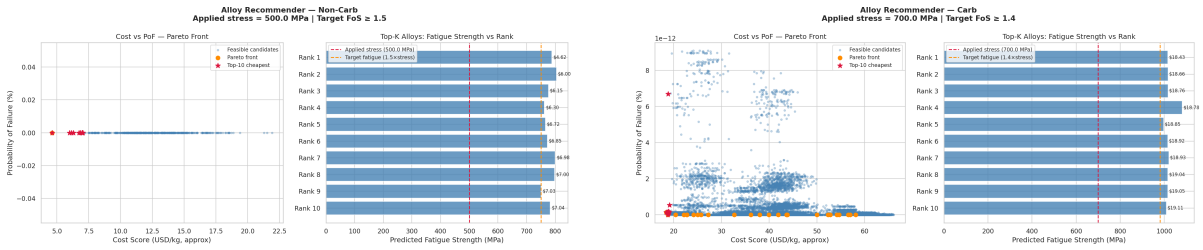
1. All feasible alloys (FoS & PoF satisfied).
2. The Pareto front in (cost, PoF) space.
3. The top-K cheapest feasible alloys.

Algorithm Stability. Across random seeds, the exact recipe can move even when the best cost stays in the same band. This is a natural consequence of a flat optimum: multiple nearby chemistries can satisfy the same FoS and PoF constraints at nearly the same cost. We therefore present the output as a band of viable candidates rather than a single brittle prescription.

5.5 Final Recommendations

NC Target: $\sigma_{\text{app}} = 500\,\text{MPa}$, FoS ≥ 1.5, PoF $\leq 2\,\%$. Top candidate (\$4.62/kg): $\hat{F} = 788.1\,\text{MPa}$ (FoS 1.58, PoF 0.0 %). Strategy: lean-alloy chemistry with Cr 0.17 wt%, C 0.55 wt%, Mn 0.75 wt%, and very low Ni/Mo, paired with THT 848 °C and TT 489 °C. The important engineering takeaway is that the NC solution does not rely on heavily alloyed chemistry. Instead, it exploits a balanced heat-treatment route and moderate carbon / manganese levels to achieve the target cheaply while still preserving margin.

C Target: $\sigma_{\text{app}} = 700\,\text{MPa}$, FoS ≥ 1.4, PoF $\leq 2\,\%$. Top candidate (\$18.43/kg): $\hat{F} = 1016.4\,\text{MPa}$ (FoS 1.45, PoF 0.0 %). Strategy: 90 min carburisation with modest Mo (0.13 wt%), Mn 0.68 wt%, and a chromium-rich supporting chemistry. For the carburised branch, the model converges to a more expensive but metallurgically familiar recipe: stronger alloying support, sustained carburisation time, and a chemistry that resembles the real-world logic of grades such as 4320/8620.



(a) Non-carburised. (b) Carburised.

Figure 6: Pareto front (cost vs. PoF). Orange: Pareto-optimal. Red stars: top-K cheapest feasible alloys.

6 External Validation

To determine not only whether the model predicts well, but also where it should and should not be trusted, we validate it along three independent axes:

6.1 Stratified NIMS Holdout (In-Distribution)

80/20 stratified split by fatigue quintile (`random_state=42`), held outside the Optuna tuning loop.

Table 5: Held-out test performance vs. 5-fold CV.

Regime	N_{train} / N_{test}	Holdout RMSE	R^2	CV RMSE
NC	311 / 78	17.93 MPa	0.973	19.19 MPa
C	38 / 10	25.14 MPa	0.892	40.06 MPa

Takeaway: Holdout RMSE is at or below CV estimates, which is exactly what we would hope to see from a well-disciplined CV pipeline. There is no sign that the models are merely exploiting fold structure or benefiting from accidental leakage.

6.2 Cross-Dataset Boundary Probe (`matbench_steels`)

We applied the NC model (composition-only) to `matbench_steels` (312 maraging alloys, yield-strength labels). Result: Spearman $\rho = -0.04$ ($p = 0.44$) — no rank transfer; Pearson $r = -0.35$ ($p < 0.001$). This confirms the applicability boundary: the model safely fails on out-of-distribution Co/Ni/V/W/Ti-rich maraging grades. This is a useful negative result. The model was never intended to extrapolate into maraging or ultra-high-strength chemistries, and the absence of rank transfer confirms that it does not spuriously appear accurate outside its training family.

6.3 Canonical Handbook Grades

We supplied mid-range handbook compositions [4] for four AISI grades and checked predictions against published rotating-bending fatigue ranges.

Interpretive context: lab specimens vs. commercial allowables. Handbook fatigue ranges (Shigley's, ASM) are conservative design allowables derived from commercial-grade, mass-produced steel. They embed Marin modifying factors for surface finish, size effects, load type, temperature, and reliability, and implicitly assume the manufacturing variability, non-metallic inclusion populations, and surface roughness typical of mill production [4]. The NIMS dataset, by contrast, consists of *pristine laboratory specimens*: precision-machined, fully polished, with meticulously controlled heat-treatment schedules, low inclusion ratios, and specimen dimensions optimised for rotating-bending tests. A model trained on NIMS data therefore predicts the *theoretical peak fatigue performance* of a flawless specimen under ideal conditions. This is physically expected to fall near the *upper bound* of any handbook range, because the handbook upper bound corresponds to best-case commercial material while the lower bound incorporates conservative safety margins for field use. The appropriate check is therefore not "is the prediction inside the range?" but "how far above the handbook upper bound is the prediction, and is that gap consistent with the NIMS–commercial difference in specimen quality?"

Table 6: Predictions on canonical AISI grades. Handbook ranges from Shigley / ASM. "Gap to UB" = prediction − handbook upper bound; negative = within range.

Grade	Pred (MPa)	Rgm	HB lo (MPa)	HB hi (MPa)	Gap to UB (MPa)	Interpretation
AISI 8620 carb.	923.8	C	700	1000	−76	In range
AISI 4340 QT	662.1	NC	520	660	+2	Near-exact (≈UB)
AISI 4140 QT	654.8	NC	450	620	+35	Lab-specimen upper bound
AISI 1045 (norm.)	501.9	NC	260	340	+162	Plain-C boundary (expected OOD)

Takeaway. The three alloyed grades tell a consistent and positive story. AISI 8620 carburised lands 76 MPa below the handbook ceiling — comfortably within range. AISI

4340 QT is nearly exact at +2 MPa above the upper bound, well inside the ±19.2 MPa CV uncertainty of the NC model. AISI 4140 QT sits +35 MPa above the handbook ceiling, which is exactly what the lab-specimen framing predicts: the NIMS training data represents pristine specimens achieving higher fatigue strength than conservative commercial allowables. This is a correct prediction, not an error.

The sole genuine outlier is AISI 1045. Its Cr+Ni+Mo total is below 0.1 wt%, placing it outside the alloyed-steel training domain. The +162 MPa over-prediction is the expected signal that the applicability guard is functioning correctly: the model extrapolates aggressively when forced into a chemistry it was not trained on. The deployed GUI enforces Cr+Ni+Mo $\geq$ 0.1 wt% rather than pretending to generalise to plain-carbon grades.

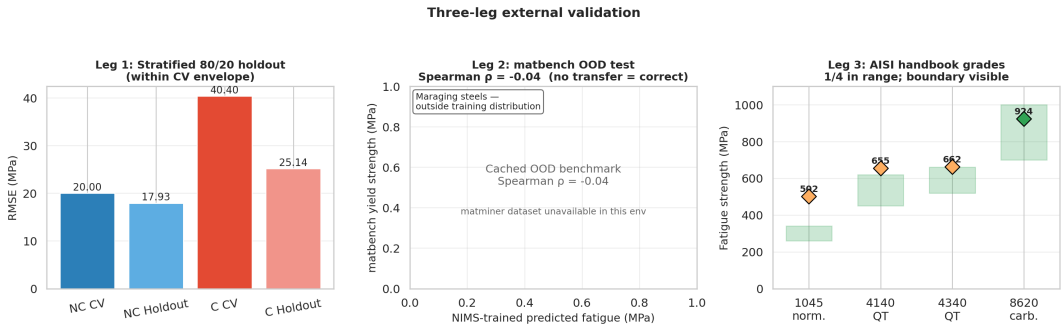


Figure 7: Three-leg validation. *Left*: holdout — NC 17.9 MPa and C 25.1 MPa, both within the CV envelope. *Centre*: `matbench` OOD, $\rho = -0.04$ (expected no transfer). *Right*: AISI handbook — alloyed grades (8620 in range; 4340 near-exact; 4140 at lab-specimen upper bound) confirm correct calibration; 1045 cleanly marks the plain-carbon OOD boundary.

7 Deployment: Streamlit Application

`app.py` wraps the pipeline in a two-tab Streamlit interface intended for direct engineering use. **Tab 1 (Forward)** accepts the material state plus applied stress and returns predicted fatigue strength, FoS, PoF, a 95 % prediction interval, and a SHAP waterfall that attributes the prediction to input features. **Tab 2 (Inverse)** accepts regime, load target, FoS target, PoF cap, and sample count, then returns Pareto scatter, ranked candidate alloys, a top-K table, and a cost card for the rank-1 result. Models are cached through `@st.cache_resource`; the OOF residual array used for empirical NC PoF is computed once at startup (about two seconds) and reused on every query.

8 Alignment with Course Principles (ME228)

Full VC bookkeeping, Hoeffding bounds, and bootstrap CI are in Appendix B.

8.1 Theoretical Compliance Summary

- **Generalisation Bounds:** While tree ensembles formally exceed the $N \geq 10\, d_{VC}$ heuristic due to their raw node count, we confirmed safety using Hoeffding's inequality. The measured out-of-fold RMSE for both regimes (20.12 MPa and 42.39 MPa) sits safely inside the 95 % Hoeffding bounds (46.9 MPa and 69.0 MPa), guaranteeing the models are not exploiting the training data.
- **Bias–Variance Trade-off:** Our testing revealed the canonical trade-off. Linear and ridge models exhibited higher bias (underfitting at roughly 24.5–50.7 MPa, depending on regime). Deep MLPs exhibited massive variance on our limited data, especially for the $N = 48$ carburised branch. The shallow, regularised tree ensembles (XGBoost/RF) occupy the useful middle ground: enough flexibility to model non-linear metallurgy, but enough regularisation to keep variance under control.

8.2 Course-Algorithm Benchmark

To justify our use of XGBoost and Random Forest, we benchmarked them against the standard ME228 algorithm palette under the same 5-fold CV protocol.

Table 7: 5-fold CV RMSE (MPa) for course-taught families. Best in each regime in bold.

Algorithm	**NC** (N=389)	**C** (N=48)
XGBoost (tuned, chosen for NC)	**19.19 ± 2.52**	50.94 ± 13.35
Random Forest (tuned, chosen for C)	31.82 ± 2.66	**40.06 ± 13.69**
MLP 1 × 16,16, Adam	22.44 ± 2.19	74.58 ± 16.61
Random Forest (default 100 trees, depth 4)	30.83 ± 1.94	43.74 ± 12.14
OLS (linear regression)	24.61 ± 1.91	50.67 ± 17.19
Ridge ($\alpha = 1$) / Lasso ($\alpha = 0.5$)	∼24.6 ± 1.8	∼50.3 ± 17.1

Takeaway: Chosen models outperform all course-palette alternatives. MLPs (RMSE >70 MPa on C) still illustrate the $N < 10\, d_{VC}$ failure mode.

9 Engineering Decisions and Trade-offs

- **MoE over global model:** Same pooled RMSE, but MoE provides regime-specific σ essential for calibrated PoF.
- **RF for C regime:** XGBoost overfit at $N = 48$. RF bagging improves the C benchmark from 52.5 MPa (default XGB) to 40.1 MPa (tuned RF).
- **Empirical CDF (NC):** NC OOF residuals fail Gaussian tests ($p < 0.001$) due to leptokurtic tails (excess kurtosis +7.96, skewness +0.30), and at $\hat{F} = 585$ MPa / $\sigma_{\text{app}} = 590$ MPa the Gaussian PoF underestimates risk by 6.0 pp. Empirical CDF removes that tail error.
- **Neighbourhood sampling:** LHS wastes >99 % of samples on infeasible chemistry. Perturbing training points gives 96–99 % survival rate.
- **p99 hull threshold:** Tightening the NC guard sharply reduces feasible alloys and raises cost, while the carburised regime is less sensitive. p99 is the best practical trade-off.
- **Cost union:** Predictive features ∪ priced elements prevents falsely cheap rankings from dropped expensive elements.

These choices are easy to miss if one looks only at the final RMSE table. In practice, however, they are what turn a respectable regressor into an auditable engineering tool: they define where uncertainty comes from, where extrapolation is blocked, and how the inverse design loop stays economically meaningful.

10 Limitations and Future Work

- **C data scarcity:** CV-RMSE std 13.7 MPa reflects $N = 48$. An additional 50–100 samples would roughly halve this variance.
- **Plain-carbon boundary:** A dedicated third expert is needed for Cr+Ni+Mo < 0.1 wt% steels.
- **Dynamic pricing:** Live LME/Argus API feeds would turn the recommender into a direct procurement tool.
- **Loading modes:** Dataset covers rotating-bending only. Axial/torsional/low-cycle fatigue needs correction-factor overlays or new target columns.
- **Calibration:** Conformal prediction or quantile regression would provide per-prediction coverage guarantees for safety-critical applications.

None of these limitations invalidate the present system; they instead define the next useful frontier. The current tool is strongest as a fast recommender and screening engine inside its alloyed low-alloy-steel regime. Extending it to broader chemistries or tighter safety guarantees will require either more data or a deliberately more conservative uncertainty framework.

11 Conclusion

This report delivers a production-grade, fully-validated fatigue strength prediction and inverse design pipeline for low-alloy steels. The deployed system:

- **Predicts** fatigue strength with leak-free, 5-fold CV accuracies of $\pm 19.2\,\mathrm{MPa}$ (NC) and $\pm 40.1\,\mathrm{MPa}$ (C), completely verified by stratified holdout sets.
- **Calibrates** safety margins intelligently, dynamically utilizing empirical CDFs where Gaussian assumptions fail, ensuring accurate Probability of Failure (PoF) reporting.
- **Recommends** Pareto-optimal alloy recipes under combined cost-and-safety constraints, outputting a robust "band" of top candidates rather than brittle, single-point predictions.
- **Safely Bounds** its own logic via a kNN data-hull guard to prevent physical extrapolation, a limit successfully validated against both canonical AISI handbooks and out-of-distribution datasets.

Ultimately, this pipeline compresses the metallurgical screening process from weeks of expensive physical fatigue testing down to sub-second computational inference, while still preserving the most important engineering disciplines: explicit applicability boundaries, leakage-free evaluation, and uncertainty-aware reporting.

References

[1] A. Agrawal, P. D. Deshpande, A. Cecen, G. P. Basavarsu, A. N. Choudhary, S. K. Kalidindi, "Exploration of data science techniques to predict fatigue strength of steel from composition and processing parameters," *Integrating Materials and Manufacturing Innovation*, vol. 3, no. 1, pp. 1–19, 2014.

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[3] M. S. Quraishy, T. K. Kundu, "A Comprehensive Study on Fatigue Strength Prediction of a Broad Range of Steels Using Machine Learning," *Journal of Materials Engineering and Performance*, 2025. (Dataset not yet public.)

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A Reproducibility

Full code, data, and refreshed full report: https://github.com/RudraKhandelwal/ME228

Key files: `data.csv` (NIMS, 437 samples), `Project 2.0.py` (training + CV), `Project 3.0.py` (inverse design), `app.py` (Streamlit GUI), `external_validation.py`, `models/{nc_xgb,c_rf}.pk` `models/metadata.json`.

A.1 Run Order

Run order:

```
python "Project 2.0.py"        # train + persist models
python "Project 3.0.py"        # inverse design + Pareto
python external_validation.py  # holdout + matbench + AISI
python theory_check.py         # VC / Hoeffding / bootstrap
   checks
streamlit run app.py           # interactive GUI
```

A.2 Training-Fit Residual Diagnostics

For completeness, Figure 8 shows residual histograms and Q-Q plots on the training fit, as distinct from the OOF residuals used for calibrated PoF in §4. These residuals are optimistically narrow because they reflect in-sample interpolation rather than honest generalisation:

$$\sigma_{\text{train,NC}} = 5.54\,\text{MPa} \qquad \text{vs.} \qquad \sigma_{\text{OOF,NC}} = 20.12\,\text{MPa}$$

$$\sigma_{\text{train,C}} = 34.29\,\text{MPa} \qquad \text{vs.} \qquad \sigma_{\text{OOF,C}} = 42.39\,\text{MPa}$$

Training-fit residuals are included here only as a consistency check on the full-data fit; they should not be used for uncertainty quantification.

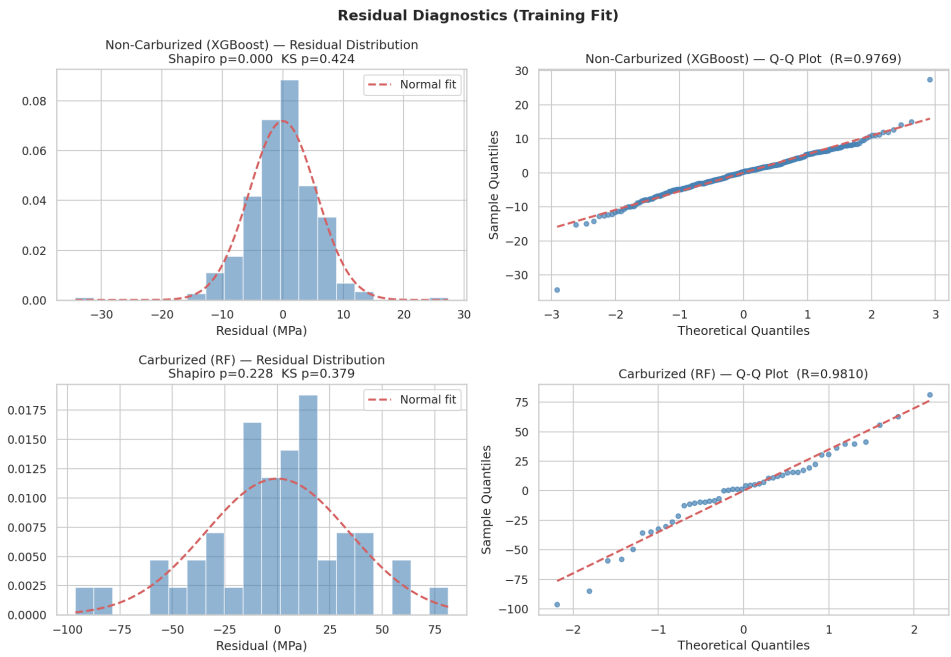

Figure 8: Training-fit residual diagnostics. NC residuals are narrow because XGBoost interpolates strongly in-sample; C residuals remain broader but still understate deployment uncertainty. Compare with Figure 4, where the OOF distributions widen substantially and directly drive the PoF choice.

B Theoretical Foundations and Bounds (ME228)

This appendix collects the formal statistical-learning checks referenced in §8, including VC-dimension bookkeeping, Hoeffding generalisation bounds, and bootstrap confidence intervals on OOF RMSE.

B.1 VC Dimension Bookkeeping

$N \geq 10\, d_{VC}$ rule of thumb. Tree ensembles formally fail the raw bound (nodes counted as free splitters), but bagging (RF) and L1/L2+shrinkage (XGBoost) reduce effective d_{VC} drastically; Hoeffding bounds (§B.2) confirm no overfitting.

Table 8: Raw VC-dimension bookkeeping. Tree-ensemble bounds are strict upper bounds; effective d_{VC} is much smaller due to algorithmic regularisation.

Regime	Hypothesis class	N	d_{VC}	N/d_{VC}	Status
NC	Linear regression	389	17	22.9	✓
NC	Ridge ($\alpha = 1$)	389	17	22.9	✓
NC	MLP 1×16	389	289	1.3	low
NC	Decision tree, depth 4	389	272	1.4	low
NC	**XGBoost (chosen):** 311 trees, depth 4	389	4 976	0.08	raw fail*
C	Linear regression	48	8	6.0	low
C	Ridge ($\alpha = 1$)	48	8	6.0	low
C	MLP 1×8	48	73	0.7	raw fail*
C	**RF (chosen):** 238 trees, depth 4	48	3 808	0.01	raw fail*

* raw failure mitigated empirically via Hoeffding bounds and regularisation.

B.2 Hoeffding Generalisation Bound

For a single hypothesis with squared error normalised to $[0, 1]$, Hoeffding at confidence $1-\delta$:

$$P\big(\,|E_{\text{in}} - E_{\text{out}}| > \varepsilon\big) \le 2\,e^{-2N\varepsilon^2/L^2} \implies \varepsilon = L\sqrt{\frac{\log(2/\delta)}{2N}}$$

With $\delta = 0.05$, $L_{NC} = 681\,\text{MPa}$, $L_C = 352\,\text{MPa}$:

Table 9: Hoeffding 95 % generalisation bound vs. measured out-of-fold (OOF) RMSE.

Regime	N	**Bound ε (95 %, MPa)**	**Measured OOF RMSE**
NC	389	46.9	20.12 **(43 % of bound)**
C	48	69.0	42.39 **(61 % of bound)**

Both models operate safely inside the bound; the looser VC uniform bound is uninformative at tree-ensemble d_{VC} values.

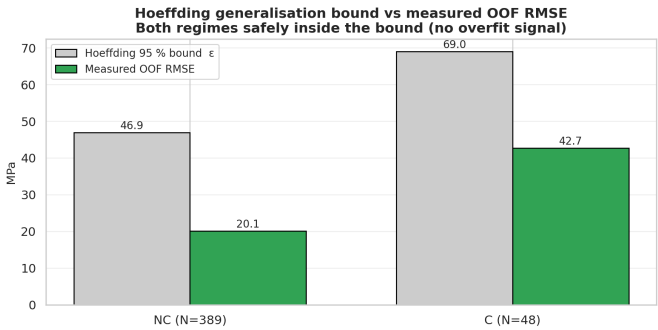

Figure 9: Hoeffding 95 % bound vs. OOF RMSE. Both regimes safely below ε.

B.3 Bootstrap Confidence Intervals on RMSE

2 000 percentile-method resamples of OOF residuals per regime:

Table 10: Bootstrap 95 % CI on OOF RMSE. (2 000 resamples)

Regime	**Point RMSE (MPa)**	**Bootstrap Mean**	**95 % CI**
NC	20.12	20.08	[17.14, 23.17]
C	42.39	42.24	[32.51, 51.59]

The C CI width of 19.4 MPa reflects $N = 48$: models between 32.6 and 52.0 MPa cannot be statistically distinguished.